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322712 (22)

BE (7th Semester)

Examination, April-May, 2013

Branch : CSE

PARALLEL PROCESSOR AND COMPUTING

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : There are five units in the question paper. In each unit first question is compulsory. Only two question of remaining three questions is to be attempted in each unit.

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Unit - I

Q. 1. Write the Moore's law.

Q. 2. Write a short note on cache coherence and MP protocols.

Q. 3. Summarize all forms of the parallelism that can be exploited at different processing levels of computer system, including both uniprocessor and multiprocessor approaches.

Q. 4. Explain the principles of scalable performance processor in brief.

Unit - II

Q. 1. Explain data and resource dependencies.

Q. 2. Write down the difference between control flow and data flow.

Q. 3. Explain the architecture of array processing

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Q. 4. Explain the principles of linear pipeling and also explain speedup, efficiency and throughput. 7

Unit - III

Q. 1. Draw a 3-cube network. 2

Q. 2. What do you mean by PRAM? Explain the type of PRAM and also explain the data broadcasting with respect to PRAM. 7

Q. 3. Write down the parallel selection algorithm and calculate the complexity. 7

Q. 4. Write the short note on following : 7

(1) Wormhole routing

(2) Hypercube architecture

Unit - IV

Q. 1. Explain hit ratio? 2

Q. 2. Explain in brief multithreading and latency hiding. 7

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Q. 3. Explain the architecture of vector-parallel Cray Y-MP. 7

Q. 4. Explain the parallel programming paralle operating system. 7

Unit - V

Q. 1. Explain Memory Hierarchy. 2

Q. 2. Explain the architecture of CC-Numa Stanford DASH. 7

Q. 3. Explain the architecture of message-passing MIMD machine. 7

Q. 4. Explain in brief MIN-Based BBN Butterfly. 7

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Q. 1. (a

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(b) What do you mean by classfull addressing? Discuss classes and blocks available with proper example. 7

(c) Describe the function of ARP and RARP with packet format. 7

(d) Discuss different types of messages sent by ICMP protocol with packet format. 7

Q. 2. (a) What do you understand by reference points ? 2

(b) Explain DIAS network architecture with example. 7

(c) Draw and explain ISDN protocol stack. Also write about ISDN focusing on ISDN principles. 7

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(d) Explain the followings : 7

(i) Signaling system

(ii) Types of switching used by ATM

Q. 3. (a) What do you mean by wireless media? 2

(b) Explain the concept of LMDS/PCS WLL. Draw block diagram also. 7

(c) Define mobility and explain direct and indirect routing approaches. 7

(d) Draw and explain the working of bluetooth security mechanism. 7

Q. 4. (a) Write down any two differences between JPEG and MPEG. 2

(b) Briefly state the Huffman coding algorithm. Show how you would use Huffman coding to encode the following set of tokens? 7

AAABDCEFBBAADCDF

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(c) Explain the operation of basic DPCM signal encoder and decoder. Also explain the source of errors that can arise. 7

(d) What are the basic of image compression? Explain JPEG compression. 7

Q. 5. (a) Define the video-on-demand system (VOD). 2

(b) What is MMX technology instruction? Also explain the data type and instruction set of MMX technology instruction. 7

(c) Write short notes : 7

(i) Content based image retrieval

(ii) IEEE 1394 interface

(d) Describe the features of MPEG-7. 7

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